

Study Session Notes 02/12/2026

11) A linear time-invariant (LTI) system with impulse response $h(t) = 2e^{-t}u(t)$ has input $x(t) = \cos(4t)\delta(t - \pi/4)$. The output is _____.

- a. $-2e^{(\pi/4-t)}u(t - \pi/4)$
- b. $-2e^{(t-\pi/4)}u(t - \pi/4)$
- c. $-2e^{(\pi/4-t)}u(t - \pi/4)$
- d. $-2e^{-t}u(t)$

$$\int_{-\infty}^{\infty} \cos(4z) \delta(z - \frac{\pi}{4}) 2e^{-(t-z)} u(t-z) dz$$

shifting whole thing to right +

$$\int_{\frac{\pi}{4}}^{\infty} 2e^{-(t-z)} u(t-z) dz$$

$$= u(t - \frac{\pi}{4}) \int_{\frac{\pi}{4}}^t \cos(4z) 2e^{-(t-z)} dz = 2u(t - \frac{\pi}{4}) \int_{\frac{\pi}{4}}^t \cos(4z) e^{t-z} dz$$

$$= 2u(t - \frac{\pi}{4}) e^t \int_{\frac{\pi}{4}}^t \cos(4z) e^{-z} dz$$

$$= 2u(t - \frac{\pi}{4}) e^t \left(\frac{z \sin(4z)}{4} - \int \sin(4z) z dz \right)$$

$$= 2u(t - \frac{\pi}{4}) e^t \left(\frac{z \sin(4z)}{4} - \frac{1}{4} \left(-\frac{z^2 \cos(4z)}{4} + \int \frac{z \cos(4z)}{4} dz \right) \right)$$

$$= 2u(t - \frac{\pi}{4}) e^t \left(\frac{z \sin(4z)}{4} + \frac{z^2 \cos(4z)}{16} - \frac{1}{16} \int z^2 \cos(4z) dz \right)$$

$$= 2u(t - \frac{\pi}{4}) e^t \left(\frac{z \sin(4z)}{4} + \frac{z^2 \cos(4z)}{16} - \frac{1}{16} \left(\frac{z^3 \sin(4z)}{4} + \frac{3z^2 \cos(4z)}{16} - \frac{3}{16} \int z \cos(4z) dz \right) \right)$$

$$= \frac{32}{33} u(t - \frac{\pi}{4}) e^t \left(\frac{z \sin(4z)}{4} + \frac{z^2 \cos(4z)}{16} \right)$$

$$y(t) = \int_{-\infty}^{\infty} x(z) h(t-z) dz$$

Friday Plan:

1 hr functions
 • Delta
 • Ramp, etc
 2 hr WS4

1 hr past exams

Saturday: All past exams

$$uv - \int v du$$

$$v = e^z \quad dv = \cos(4z)$$

$$du = e^z dz \quad v = \frac{\sin(4z)}{4}$$

$$= 2u(t - \frac{\pi}{4}) e^t \int_{\frac{\pi}{4}}^t z \cos(4z) dz$$

$$= \left(\frac{32}{16} + \frac{1}{16} \right) u(t - \frac{\pi}{4}) \int_{\frac{\pi}{4}}^t z \cos(4z) dz$$

$$v = e^z \quad dv = \sin(4z)$$

$$du = e^z dz \quad v = -\frac{\cos(4z)}{4}$$

Study Session Notes 2.

$$= \frac{32}{33} u(t - \frac{\pi}{4}) e^{\frac{t}{4}} \left(\frac{e^{\frac{\pi}{4}} \sin(4 \cdot \frac{\pi}{4})}{4} \Big|_{\frac{\pi}{4}}^t + \frac{e^{\frac{\pi}{4}} \cos(4 \cdot \frac{\pi}{4})}{16} \Big|_{\frac{\pi}{4}}^t \right)$$

$$= \frac{128}{33} u(t - \frac{\pi}{4}) e^{\frac{t}{4}} \left(\frac{e^{\frac{\pi}{4}} \sin(4t)}{4} \Big|_{\frac{\pi}{4}}^t + \frac{e^{\frac{\pi}{4}} \cos(4t)}{16} \Big|_{\frac{\pi}{4}}^t \right)$$

$$= \frac{128}{33} u(t - \frac{\pi}{4}) e^{\frac{t}{4}} \left[\left(\frac{e^{\frac{\pi}{4}} \sin(4t)}{4} - \frac{e^{\frac{\pi}{4}} \sin(4 \cdot \frac{\pi}{4})}{4} \right) + \left(\frac{e^{\frac{\pi}{4}} \cos(4t)}{16} - \frac{e^{\frac{\pi}{4}} \cos(4 \cdot \frac{\pi}{4})}{16} \right) \right]$$

$$\int_{-\infty}^{\infty} \cos(4z) \delta(z - \frac{\pi}{4}) 2e^{-(t-z)} u(t-z) dz$$

$$\left(\cos(4t) \delta(t - \frac{\pi}{4}) = \cos(4(t - \frac{\pi}{4})) \Rightarrow \text{based on time-shift property} \right)$$

$$2u(t - \frac{\pi}{4}) \int_{\frac{\pi}{4}}^t \cos(4(z - \frac{\pi}{4})) e^{-(t-z)} dz$$

$$= 2u(t - \frac{\pi}{4}) \int_{\frac{\pi}{4}}^t \cos(4z - \pi) e^{-(t-z)} dz$$

$$= 2u(t - \frac{\pi}{4}) e^{-t} \int_{\frac{\pi}{4}}^t \cos(4z - \pi) e^z dz$$

EASY WAY! (Ignore the other stuff.)

1. Use Sampling Theorem

Sampling property of $\delta(t)$

As was noted earlier, multiplying an impulse function by a constant k gives a scaled impulse of area k . Now we consider what happens when a continuous-time function $x(t)$ is multiplied by $\delta(t)$. Since $\delta(t)$ is zero everywhere except at $t = 0$, it follows that

$$x(t)\delta(t) = x(0)\delta(t),$$

provided that $x(t)$ is continuous at $t = 0$. By extension, multiplication of $x(t)$ by the time-shifted impulse function $\delta(t - T)$ gives

$$x(t)\delta(t - T) = x(T)\delta(t - T).$$

$$x(t) \delta(t - T) = \cos(4t) \delta(t - \frac{\pi}{4}) \Rightarrow \cos(4 \cdot \frac{\pi}{4}) \delta(t - \frac{\pi}{4})$$

$$= -\delta(t - \frac{\pi}{4})$$

6. Convolution with Impulse $x(t) * \delta(t - T) = x(t - T)$

$$x(t) * \delta(t - T) = -\delta(t - \frac{\pi}{4}) \cdot 2e^{-t} u(t) = \boxed{-2e^{-(t - \frac{\pi}{4})} u(t - \frac{\pi}{4})}$$

Study Session Notes cont. 3

12) $x(t) = tu(t)$ and $y(t) = \delta(t) + \delta(t-1)$. The convolution $z(t) = x(t) * y(t) =$ _____.

- a. $u(t)$
- b. $tu(t) + (t-1)u(t-1)$**
- c. $(2t-1)u(t)$
- d. $t-1$

$$x(t) * h_1(t) + x(t) * h_2(t) \quad \text{Distributive Property}$$

$$= t u(t) + (t-1) u(t-1) \quad \text{Convolution w/ Impulse}$$

13) The convolution $u(t) * \delta(t-1) * u(t) =$ _____.

- a. $u(t-1)$
- b. $tu(t-1)$
- c. $(t-1)u(t)$
- d. $(t-1)u(t-1)$**

$$u(t) * \delta(t-1) = u(t-1) \quad \text{Convolution w/ impulse}$$

$$u(t-1) u(t) = u(t-1) \int_1^t 1 dz = u(t-1)(t-1) \quad \text{Universal Integrator}$$

Given $\int_{-\infty}^{\infty} x(t) dt = 2$ and $\int_{-\infty}^{\infty} y(t) dt = 4$ The area $\int_{-\infty}^{\infty} z(t) dt$ under the convolution signal $z(t) = x(t) * y(t)$ is _____.

- a. 6
- b. 8**

$$\text{Area of } y(t) = \text{Area } x(t) * \text{Area } h(t)$$

$$= 2 \times 4 = 8$$

Convolution integral: $y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$

Causal systems and signals: $y(t) = h(t) * x(t) = u(t) \int_0^t h(\tau) x(t-\tau) d\tau$

Property	Description
1. Commutative	$x(t) * h(t) = h(t) * x(t)$
2. Associative	$[g(t) * h(t)] * x(t) = g(t) * [h(t) * x(t)]$
3. Distributive	$x(t) * [h_1(t) + \dots + h_N(t)] = x(t) * h_1(t) + \dots + x(t) * h_N(t)$
4. Causal * Causal = Causal	$y(t) = u(t) \int_0^t h(\tau) x(t-\tau) d\tau$
5. Time-shift	$h(t-T_1) * x(t-T_2) = y(t-T_1-T_2)$
6. Convolution with Impulse	$x(t) * \delta(t-T) = x(t-T)$
7. Width	Width of $y(t)$ = width of $x(t)$ + width of $h(t)$
8. Area	Area of $y(t)$ = area of $x(t)$ * area of $h(t)$
9. Convolution with $u(t)$	$y(t) = x(t) * u(t) = \int_{-\infty}^t x(\tau) d\tau$ (Ideal integrator)
10a. Differentiation	$\left(\frac{d^m x}{dt^m}\right) * \left(\frac{d^n h}{dt^n}\right) = \frac{d^{m+n} y}{dt^{m+n}}$
10b. Integration	$\int_{-\infty}^t y(\tau) d\tau = x(t) * \left[\int_{-\infty}^t h(\tau) d\tau\right] = \left[\int_{-\infty}^t x(\tau) d\tau\right] * h(t)$

SUMMARY OF BASIC TRICKS

$$x(t) * \delta(t-T) = x(t-T) \quad \text{DELAYER}$$

$$x(t) * u(t) = \int_{-\infty}^t x(\tau) d\tau \quad \text{UNIVERSAL INTEGRATOR}$$

$$u(t) * u(t) = r(t) \quad \text{INTEGRATE THE UNIT STEP}$$

$$r(t) * u(t) = \frac{1}{2} t^2 u(t) \quad \text{INTEGRATE THE RAMP}$$

Study Notes cont. 4

15) Given $e^{-t}u(t) * e^{-t}u(t) = te^{-t}u(t)$ and $e^{-t}u(t) * u(t) = (1 - e^{-t})u(t)$

The convolution $e^{-t}u(t) * [2u(t) + e^{-t}u(t)]$ is _____.

- a. $[(t-2)e^{-t} + 2]u(t)$
- b. $(t-2)e^{-t} + 2$
- c. $(te^{-t} + 2)u(t)$
- d. $2[(t-1)e^{-t} + 1]u(t)$

$$\begin{aligned}
 & e^{-t}u(t) * [2u(t) + e^{-t}u(t)] \\
 &= e^{-t}u(t) * 2u(t) + e^{-t}u(t) * e^{-t}u(t) \quad \text{Distributive Property} \\
 &= 2(1 - e^{-t})u(t) + te^{-t}u(t) \\
 &= 2u(t) - 2e^{-t}u(t) + te^{-t}u(t) = e^{-t}(tu(t) - 2u(t)) + 2u(t) \\
 &= e^{-t}u(t)(t-2) + 2u(t) \\
 &= u(t)(e^{-t}(t-2) + 2)
 \end{aligned}$$

16) The convolution $e^{-t}u(t) * e^{-2t}u(t) = (e^{-t} - e^{-2t})u(t)$. Using the time-shift property, the convolution $e^{2-t}u(t-2) * e^{2-2t}u(t-1)$ is _____.

- a. $(e^{2-t} - e^{2-2t})u(t-2)$
- b. $(e^{2-t} - e^{2-2t})u(t-3)$
- c. $\frac{1}{3}(e^{2-t} - e^{2-2t})u(t-3)$
- d. $(e^{3-t} - e^{6-2t})u(t-3)$

$$-(t-2) = 2-t$$

$$-(t-2) =$$

$$t - T_1 - T_2$$

$$\begin{aligned}
 &= \frac{-(t-1-2)}{(e^{3-t} - e^{6-2t})}u(t-3) = (e^{3-t} - e^{6-2t})u(t-3) \\
 &= (e^{3-t} - e^{6-2t})u(t-3)
 \end{aligned}$$

Study Session Notes cont. 5

17) The convolution $\cos(3t) * \delta(t - \pi/3) = \underline{\hspace{2cm}}$.

a. $-\cos(3t)$

b. $\cos(3t - \pi/3)$

c. -1

d. $\sin(3t)$

convolution w/ impulse $\Rightarrow \cos(3t) * \delta(t - \frac{\pi}{3}) = \cos(3(t - \frac{\pi}{3}))$
 $= \cos(3t - \pi)$

$\cos(A - B) = \cos A \cos B + \sin A \sin B$

$\Rightarrow \cos(3t - \pi) = \cos(3t) \cos(\pi) + \sin(3t) \sin(\pi)$
 $= -\cos(3t)$

18) The convolution $x(t) * y(t) = e^{-2t}$. The convolution of the derivatives $\frac{dx}{dt} * \frac{dy}{dt} = \underline{\hspace{2cm}}$.

a. $4t^2 e^{-2t}$

b. $4te^{-2t}$

c. $-2e^{-2t}$

d. $4e^{-2t}$

10a. Differentiation

$$\left(\frac{d^m x}{dt^m} \right) * \left(\frac{d^n h}{dt^n} \right) = \frac{d^{m+n} y}{dt^{m+n}}$$

$\Rightarrow$ take 2nd derivative $\Rightarrow 4e^{-2t}$